Project Report: Mercedes-Benz Greener Manufacturing

1. Project Overview

The objective of this project was to develop a robust machine learning algorithm to predict the time a Mercedes-Benz car spends on the test bench given various customized feature configurations. By accurately predicting testing time, the algorithm assists in optimizing the testing system, resulting in faster testing processes and lower carbon dioxide emissions.

2. Data Ingestion and Validation

The initial phase involved loading and inspecting the provided datasets:

- **Data Loading:** The training dataset (train.csv) and testing dataset (test.csv) were successfully loaded into Pandas DataFrames. The initial shape of the training data was 4,209 rows and 378 columns, while the test data contained 4,209 rows and 377 columns.
- **Null Value Check:** Both datasets were evaluated for missing values. It was confirmed that there were zero missing (null) values in both the training and testing sets.
- **Unique Values:** The cardinality of the columns was verified to understand the distribution of unique values across the dataset.

3. Data Preprocessing

To prepare the data for the machine learning algorithm, two primary cleaning and transformation steps were executed:

- **Zero Variance Feature Removal:** Numerical columns were analyzed for variance. Features with a variance exactly equal to zero (meaning every row contains the exact same value) carry no predictive power and were removed to reduce computational noise. A total of 12 features were successfully identified and dropped from both datasets (specifically: 'X11', 'X93', 'X107', 'X233', 'X235', 'X268', 'X289', 'X290', 'X293', 'X297', 'X330', and 'X347').
- **Categorical Encoding:** Machine learning models require numerical inputs. Categorical features (represented as 'object' data types) were transformed using LabelEncoder. The encoder was fit on a combined set of both training and testing categorical data to ensure all possible unique string labels were accurately recognized and converted into integer formats.

4. Dimensionality Reduction

Due to the high dimensionality of the anonymized feature space (364 features after dropping the ID and target variable), Principal Component Analysis (PCA) was applied.

- **Transformation:** The dataset was condensed into 12 principal components (n_components=12). This step effectively minimized multi-collinearity and reduced the feature space while retaining the most critical variance required for predictive modeling.

5. Model Training and Prediction

With the data cleaned, encoded, and reduced, a regression model was built to predict the target variable y (testing time).

- **Algorithm Selection:** An XGBoost Regressor (XGBRegressor) was chosen for its high performance and efficiency with tabular data.
- **Hyperparameters:** The model was initialized with an objective of reg:squarederror, a max_depth of 4, a learning_rate of 0.1, and 150 estimators (n_estimators=150). A random_state of 42 was set to ensure reproducibility.
- **Execution:** The XGBoost model was trained on the PCA-transformed training features and the target variable. Following successful training, the model predicted the test bench times for the transformed test dataset.

6. Deliverables Generated

The final predictions were successfully mapped to their corresponding test IDs and exported as a clean CSV file named **mercedes_xgboost_submission.csv**. This file, alongside the Python script/Jupyter Notebook and the execution screenshots, comprises the final deliverables for this course-end project.